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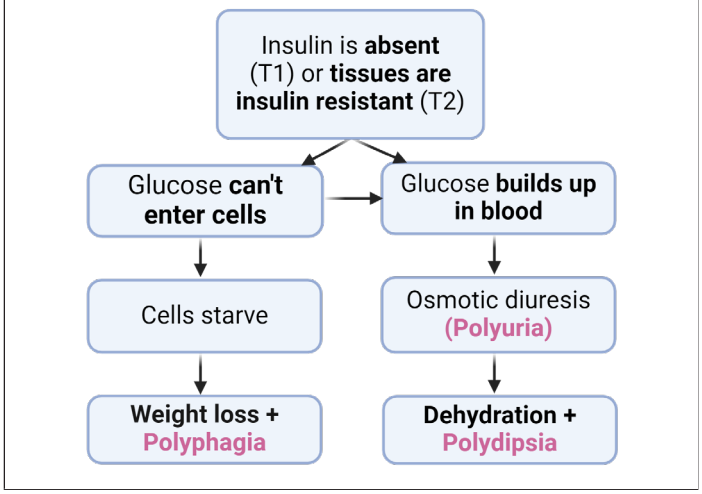
Gestational Diabetes

1. Pathophysiology

Diabetes mellitus = impaired glucose metabolism.

- Glucose can't reach cells → **Cells starve + Glucose builds up** in blood

FIGURE 1. DIABETES



- Diabetes mellitus is classified as **type 1** (T1DM), **type 2** (T2DM), or **gestational diabetes** (GDM) (see DIABETES CHEAT SHEET).
- **GDM**:
  - Higher levels of placental hormones (**estrogen**, **progesterone**) and **cortisol** in the 2nd and 3rd trimester → ↑ **insulin resistance** → ↑ **circulating glucose (hyperglycemia)**
  - Usually **resolves postpartum**

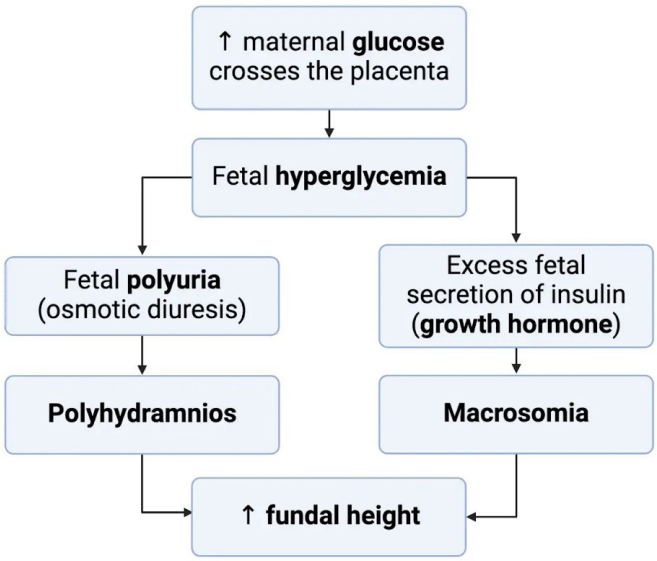
2. Risk Factors

- **Overweight**
- **History** of GDM or **signs of GDM** in previous pregnancy (macrosomia)
- **Maternal age** >25 years
- **High-risk ethnic groups** (Hispanic, Black)
- **First-degree relative** with diabetes

3. Clinical Manifestations

- **Hyperglycemia**:
  - **3 "Ps"**: Polyuria, polyphagia, polydipsia
- ↑ **fundal height** from fetal **macrosomia** and **polyhydramnios** (excess amniotic fluid):

FIGURE 2. INCREASED FUNDAL HEIGHT



4. Diagnostic Studies

- Clients are **screened** for GDM between **24-28 weeks gestation**.
- GDM is diagnosed if:
  - Positive screening result for **two-step oral glucose challenge test (GCT)** (FIGURE 3)

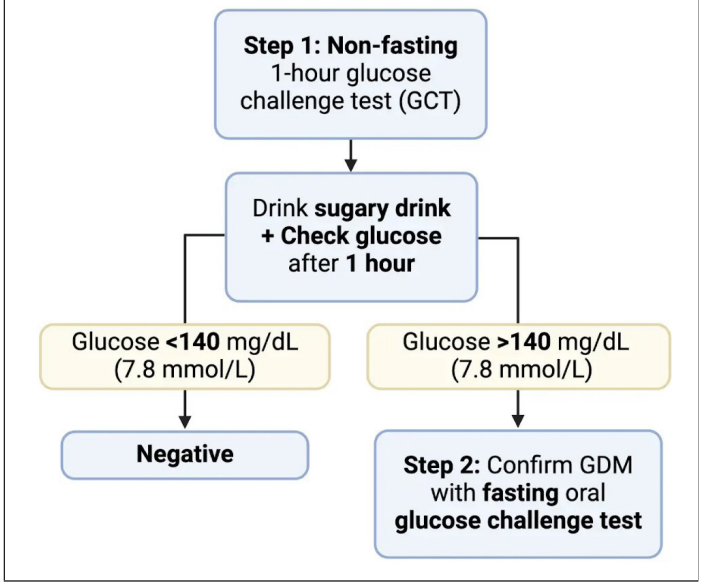
NCLEX Star Points

1. **GDM screening**: Screen for GDM between **24-28 weeks** gestation using a **non-fasting** 1-hour **oral glucose challenge test (GCT)**. If 1-hour GCT is positive, a **fasting 2-hour CGT** is performed.

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4. Diagnostic Studies, Continued

FIGURE 3. SCREENING FOR GESTATIONAL DIABETES



5. Interventions & Client Teaching

The **#1 priority** for clients with diabetes during pregnancy = maintaining **tight glycemic control** to ensure **maternal and fetal well-being** and prevent complications like spontaneous abortion and neural tube defects.

Interventions for GDM focus on:

1. Teaching **diabetes management**
2. Monitoring **maternal and fetal well-being**
3. Managing **birth and newborn complications**

1. Teach **diabetes management**:
   - **Diet and exercise**:
     - **GDM** is treated with **diet and exercise first**, then insulin as needed.
     - **Exercise 30-60 min** daily.
     - Eat a balanced diet:
       - Divide daily caloric needs between **3 meals** and **2-3 snacks (never skip meals)**.
       - Eat **high-fiber**, complex carbohydrates (brown rice) and **limit simple sugars** (white bread, candy).
       - Consult with a **registered dietician**.
   - **Blood sugar management**:
     - Blood glucose levels are monitored to evaluate whether blood sugar can be **controlled** with **diet and exercise alone**.
     - Teach client to **monitor** blood **glucose** as prescribed (usually four times daily):
       - Measure **post-prandial** glucose **1-2 hr** after eating.
     - If **hyperglycemia persists** despite healthy diet and exercise:
       - **Insulin** may become necessary for **T2DM and GDM**.
       - ↑ doses of **insulin** will be required during pregnancy for clients who have **T1DM**.
     - Monitor for **signs of hypoglycemia** (shaky, clammy, confused).
     - Treat **hypoglycemia** by following the **"Rule of 15"**: **15 grams** of carbohydrates + Recheck glucose in **15 min** (see **DIABETES CHEAT SHEET**).
2. Monitor **maternal** and **fetal well-being**:
   - Uncontrolled **hyperglycemia** in pregnancy can cause several maternal and fetal **complications** (**FIGURE 4**).
   - **Fetal** complications:
     - **Spontaneous abortions**
     - Congenital malformations (**neural tube defects**)
     - **Intrauterine growth restriction** (IUGR)
     - **Macrosomia** (newborn weight ≥4,000 g [8 lbs 13 oz])
   - **Maternal** complications:
     - **Preeclampsia** (BP ≥140/90, vision changes, headaches)
     - Frequent **UTIs**: Glucosuria ↑ bacterial growth
   - Assess for complications:
     - **Monitor** blood **glucose** as prescribed.
     - Perform early and frequent **fetal surveillance**:
       - **Ultrasound** measurements to assess fetal growth and amniotic fluid volumes
       - **Non-stress tests** and **biophysical profiles** to assess fetal well-being

NCLEX Star Points

❷ **Lifestyle changes**: Teach clients with GDM to manage blood glucose with **diet and exercise**. Divide daily calorie needs between **3 meals** and **2-3 snacks**.

5. Interventions & Client Teaching, Continued

3. Manage birth and newborn **complications**:
   - During labor, **strict glycemic control** is necessary to ↓ **risk** of **neonatal hypoglycemia** (**FIGURE 4**).
     - Monitor maternal **blood glucose** levels **every hour**.
     - Administer **IV fluids** with **dextrose** (D5NS) and **insulin** as needed to maintain **normal glucose levels** during labor.
   - Clients with **GDM** are at risk for birth **complications of macrosomia**:
     - ✪ **Postpartum hemorrhage** (PPH): Uterine overdistention = ↑ risk for uterine atony → PPH
     - **Birth injuries** (perineal lacerations, shoulder dystocia)
   - After birth, assess **newborn blood glucose** and monitor for **complications** due to GDM:
     - **Assess** newborn **blood glucose** per facility policy:
       - ✪ Place **heel warmer** on newborn to **improve blood flow** and prevent **false low** results.
       - ✪ Obtain blood sample from the **lateral heel**.

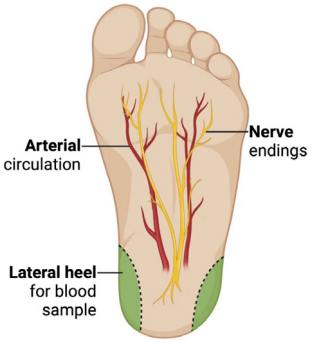
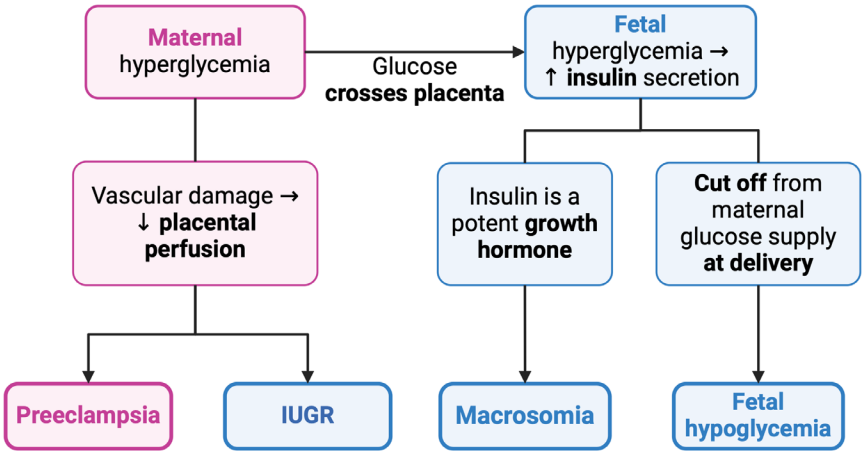


FIGURE 4. COMPLICATIONS OF GDM



   - **Hypoglycemia = <45 mg/dL** (2.5 mmol/L) in the first 72 hr of life
     - ✪ If newborn is hypoglycemic and **alert**, give **oral feedings** (breastfeed).
     - ✪ If newborn cannot feed or hypoglycemia persists, treat with **IV dextrose** instead.
   - **Newborn complications**:
     - **Hypoglycemia**:
       - ✪ **Persistent jitters** or tremors
       - Hypothermia
     - ✪ **Respiratory distress**: Insulin inhibits **surfactant** production in the fetal lungs:
       - Monitor newborn for **grunting**, **tachypnea**.

NCLEX Star Points

❸ **Risk for preeclampsia and PPH**: Clients with **GDM** are at increased risk for **preeclampsia** and **PPH**. **Monitor** the client for **uterine atony** and teach client to **report** signs of preeclampsia such as **BP ≥140/90**, **headaches**, and visual changes.

❹ **Newborn hypoglycemia symptoms**: If a **newborn** has signs of hypoglycemia, such as **persistent jitters** or hypothermia, **assess** blood **glucose** by **warming** the **heel** to promote blood flow, then collect a blood sample from the lateral heel.

❺ **Newborn hypoglycemia treatment**: If the **newborn** is hypoglycemic and **alert**, give **oral feedings**. If hypoglycemia **persists**, give **IV glucose**.

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NCLEX Top 5 Targets

1. Screen for GDM between __-__ **weeks** gestation using a _____ **(fasting or nonfasting?)** 1-hour oral **glucose challenge test**.
2. **Teach clients** diagnosed with GDM to **first manage blood sugars** with _____ and _____. Divide daily caloric needs between _____ **meals** and _____ **to** _____ **snacks**.
3. Monitor the **postpartum** client for **uterine** _____ and teach clients to **report** signs of **preeclampsia**, including **BP ≥**___/___, headaches, and _____ **changes**.
4. If the **newborn** has **persistent jitters** or hypothermia, **assess for** _____, but **first**, prevent falsely low glucose values by **warming the newborn's** _____.
5. If a newborn has **hypoglycemia** and is **alert**, treat with **oral feedings**. Treat **persistent** hypoglycemia with ____.

Answers: 1. 24-28 weeks, non-fasting 2. diet, exercise, 3, 2 to 3 3. atony, 140/90, vision 4. hypoglycemia, heel 5. IV glucose

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Attributions:

- **Heel Stick**: Modified from BioRender.com

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